COMMERCIAL ITEM DESCRIPTION

TONOMETER, OPHTHALMIC, SCHIOTZ

The General Services Administration has authorized the use of this Commercial Item Description to replace Federal Specification GG-T-585B which is being cancelled.

This commercial item description covers a mechanical tonometer suitable for measuring intraocular pressure.

Salient characteristics:

General. The tonometer furnished under this document shall be of Schiotz design using no electrical power. Scale reading is obtained in proportion to the cornea pressure. Weights are added to the instrument for expanding the scale range.

Materials.

Frame. Aluminum alloy No. 2024 or No. 6061. All tempered for the purpose intended.

Scale Support. Etched aluminum alloy.

Pointer. Spring steel or hard-drawn stainless steel.

Scale. Permanently printed on a plastic strip.

Remainder components. Bright chrome plated over brass.

Case. High impact resistant material. Form fitting pad or molded insert.

AMSC N/A FSC 6540

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:28Z
Check the source to verify that this is the current version before use.

Style, design and dimensions. The mechanical tonometer shall be Schiotz design as approved by the Committee on Standardization of Tonometers of the American Academy of Ophthalmology and Otolaryngology. The instrument shall consist of a rigid housing having a limited vertical motion within the retaining handle. There shall be a freely-moving plunger within the housing. This plunger shall carry weight or weights as described hereinafter, and shall transmit its vertical motion to a pointer by means of a lever. The motion so transmitted shall be measured by means of a scale located at the top of the instrument and over which the pointer shall pass. The scale shall be graduated from at least -1 to at least +15 in one unit steps.

Weights. Four weights shall be supplied with the instrument. One weight marked "5.5" shall be secured to the plunger by clipping action. The other three weights shall be furnished: one 2-gm weight marked "7.5", one 4.5-gm weight marked "10", and one 9.5-gm weight marked "15". The weights shall each have a tolerance of ± 0.02 gm. These weights shall be put on or off easily yet they shall be secured to the shaft.

Dimensions. Dimensions shall be in accordance with the specification 5 or latest edition for tonometer, as published by the Committee on Standardization of Tononmeters of the American Academy of Ophthalomology and Otolaryngology. Where dimensional requirements contained in this specification are in conflict with the committee's latest specification, the latter specification shall govern. Dimensions are given in millimeters (mm).

Footplate

Diameter	10.1	± 0.2
Radius of concavity	15.0	± 0.25
Diameter of Footplate hole	3.7	± 0.5

Plunger

Diameter	3.0	± 0.03
Radius of concavity of base	15.0	± 0.75
Extension of plunger below spherical surface of footplate	3 maximum	

Pointer

Depth	3 maximum
Thickness	0.25 maximum
Distance between scale and pointer	1.0 maximum

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:28Z
Check the source to verify that this is the current version before use.

Hammer

Plunger contact with hammer	smooth, free from pits throughout usable scale

Scale. Each scale unit shall equal 0.050 mm of plunger displacement. The tonometer scale shall extend from -1.00 at least + 15.0 in 1 mm increments. A test block shall be supplied in the case with each tonometer. The tonometer when resting on the convex surface of the test block shall give a reading of 0.00 ± 0.15. The scale shall be calibrated so that the plunger displacement in mm from -1.0 scale reading shall be as follows:

Scale Reading	Plunger Displacement
5.0	0.30 ± 0.01
10.0	0.55 ± 0.015
15.0	0.80 ± 0.03

Finish. The finish shall be mirror, or mirror and satin, except the pointer, which shall be given a blue or black finish. As an alternate, frame shall be fine sandblasted and anodized. Finish shall be such that it will not chip.

Chart. A chart converting scale units to mm of mercury shall be furnished with each instrument in printed form.

Case. Each instrument shall be provided with a molded insert or lined case suitably fitted to securely retain the instrument and accessories. Outside dimensions of the case shall not exceed 6-1/2 X 4-1/4 X 1-3/16 inches.

Identification marking. Each instrument shall be permanently and legibly marked with the manufacturer's name or trademark of such known character that the source of manufacture may be readily determined. Serial number shall be marked on the scale support.

Instructions for Disinfecting and Cleaning Mechanical Tonometers. The following instructions for disinfecting and cleaning mechanical tonometers as approved by the Committee on Standardization of Tonometers of the American Academy of Ophthalmology and Otolaryngology shall be furnished and packed into each instrument case:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:28Z
Check the source to verify that this is the current version before use.

Disinfection. In order to avoid the transference of infectious bacteria or viruses from one patient to another, the end of the mechanical tonometer coming in contact with the eye shall be placed in suitable stand containing a 1-5000 Benzalkonium Chloride Solution (U.S.P.) so that the mechanical tonometer footplate and plunger remain immersed in this disinfecting solution during the period of day in which the tonometer is in use. The instrument shall not be left in the disinfecting solution overnight. Since the disinfectant may be slightly irritating to the eye, the footplate and the lower end of the plunger shall be washed with distilled water and wiped dry with gauze sponge immediately before use. If a suitable stand is not available, the mechanical tonometer footplate and lower end of the plunger shall be wiped with a gauze sponge moistened with 1-5000 Benzalkonium Chloride Solution (U.S.P.), rinsed with distilled water, and wiped dry with a gauze sponge immediately before and after each use.

Cleaning. At the end of the day's use, the plunger shall be removed from the cylinder of the mechanical tonometer after removing the 5.5 gram weight. The plunger shall be wiped clean with a gauze sponge moistened with 70 to 95 percent alcohol. The plunger casing of the cylinder shall be thoroughly irrigated with warm distilled water to rapidly dissolve any crystals of salt which may have formed from tear fluid. The plunger casing shall then be irrigated with 70 to 95 percent alcohol, and the mechanical tonometer allowed to dry. The mechanical tonometer and the plunger shall then be kept in its closed carrying case until the next day's use.

Certificate of inspection and calibration. Each instrument shall be accompanied with a certificate of inspection and test certifying that the instrument is dimensionally correct and has been calibrated. The certifying laboratory shall be an independent establishment as approved by the contracting agency.

Workmanship. The mechanical tonometer shall be free from defects which detract from its appearance or impair its serviceability. There shall be no sharp or rough metal edges or corners.

Unit. Each (EA). One tonometer, as specified, constitutes one unit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:28Z
Check the source to verify that this is the current version before use.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and as specified herein. Unit of product for examination purposes shall be one tonometer. Sampling shall be level II and AQL of 1.0.

Examination. Examination shall be conducted to determine compliance with specification requirements.

Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as specified herein. Unit of product for test purposes shall be one tonometer. Sampling shall be level S-2 and AQL of 1.0.

Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:28Z
Check the source to verify that this is the current version before use.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug, and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use the covered material in accordance with Public Law 94-580, as amended, and as implemented by Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTE: The following National Stock Number is covered by this document:

NSN	ITEM IDENTIFICATION
6540-00-382-6100	TONOMETER, OPHTHALMIC, SCHIOTZ

Ordering data. Intermediate/exterior package quantities, labeling, and marking must be specified in the contract and/or order.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:28Z
Check the source to verify that this is the current version before use.

MILITARY INTERESTS:

Custodians:

ARMY - MD
NAVY - MS
AIR FORCE - 03

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS
GSA

PGC: 45760

Project No. 6540-1215

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:28Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
A-A-54044	TONOMETER, OPHTHALMIC, SCHIOTZ

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:28Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS. In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE. This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEFENSE LOGISTICS AGENCY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO 4968 WASHINGTON D C

POSTAGE WILL BE PAID BY THE DEFENSE LOGISTICS AGENCY

Commander
Defense Personnel Support Center
ATTN: DPSC- RSTE
2800 South 20th Street
Philadelphia, Pennsylvania 19101-8419

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:28Z
Check the source to verify that this is the current version before use.